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The influence of copper trade relation structure on copper price: From the perspective of industrial chain

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ABSTRACT

Trade is an important approach to coordinating the global allocation of copper. Under the changeable international situation, the volatile relation between trading nations plays an increasingly important role in copper price fluctuation. In this paper, the influence of trade relation structure on copper commodity prices is discussed by combining the complex network and econometric model. From the perspective of the industrial chain, the upstream and downstream commodities' market liquidity will cause the upstream commodity price to significantly change within four years. The midstream commodity price is mainly influenced by its market liquidity and chain-like trade relations' scale. This influence peaks around the third year. The downstream commodity price is mainly influenced by its own trade relation structure. And the influence was greatest in the second year. This article provides a reference for traders to guard against possible fluctuations in commodity prices caused by changes in relation between trading countries.

1. Introduction

Due to the unbalanced distribution of copper resources, the supply-demand of copper resources in various countries are not balanced. Trade is an effective way to realize the global allocation of copper resources and solve the imbalance between supply and demand of copper resources. Exporting countries have different prices due to their different technological development, labor force, resource reserves and foreign exchange. However, due to political, transportation and other factors, importing and exporting countries often do not decide whether to trade based solely on the quantity and price of commodities. That is, the importing country may not choose the country with the lowest import cost to import, while the exporting country may not choose the country with the most profit to export. Such trade decision-making behavior, which is not entirely based on the consideration of profit maximization, will change the trade relation structure (the embodiment of all trade relation between trading countries) of copper commodities, disturb the commodity price which originally determined by supply-demand, and then cause the copper commodity prices to fluctuate (Fig. 1).

In 2017, the United States was one of the major copper concentrate

exporters to China. However, due to the tensions between China and the United States, the trade between China and the United States almost stopped, which also led to the adjustment of the global copper trade relation structure (see Supplementary Fig. A.1). This has become an important factor in the decline of international copper prices during this period, which brings challenges to global copper enterprises. In 2019, CODELCO's chief economist mentioned that "Since the economic and trade friction in 2018, the international price of copper has fallen by about 20%. Every penny drop in the copper price per pound means that Chile has lost \$120 million in export revenue" (Finance, 2019). If traders can anticipate the possible influence of the copper commodity trade relation structure change on the price, they can develop strategies in advance to deal with such price fluctuations. This is very meaningful.

At present, many scholars have paid attention to the copper price and its influencing factors. Shao et al. (2013) studied the relation between capital speculative trading, arbitrage trading and international copper futures price fluctuations. Tang et al. (2021) discussed the influence of political risk on copper price. Hu et al. (2017) analyzed the influence of supply shock, aggregate demand shock and specific demand shock on copper price volatility. Aruga and Managi (2011) analyzed the causal relation between different copper market prices. But the trade relation

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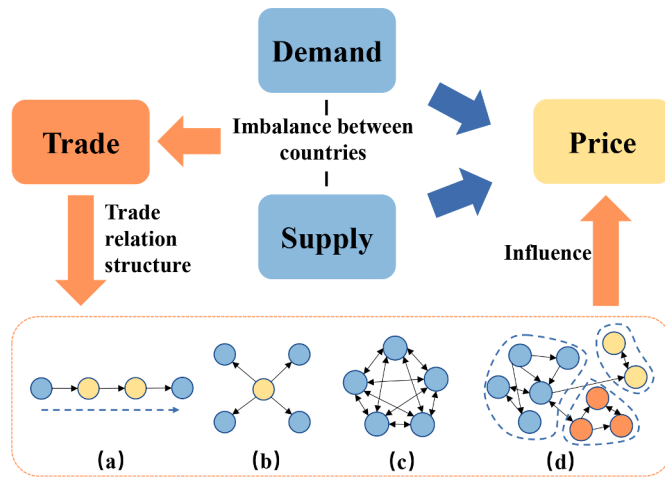


Fig. 1. The influence of trade relation structure on commodity prices.

structure, the important factor influencing copper prices, has not been analyzed. As an important resource related to national economic development and construction, the fluctuation of copper price has a very important influence on the economic development of many countries (An and Li, 2022; Namahoro et al., 2022). In the current complex international situation, the commodity trade relation structure changes frequently, which brings potential risks to copper commodity prices. This paper attempts to discuss the influence of the copper trade relation structure on copper prices, so as to provide a reference for relevant decision makers to evaluate the volatility of prices, and to better guard against the potential risks brought by the current complex international situation on copper prices.

If trading countries are regarded as nodes and the trade between them as edges, the overall trade relation structure in the world will be transformed into a trade network. In this way, the trade relation structure of a certain commodity can be quantified by selecting appropriate indicators to describe the network characteristics. At present, there are many methods to extract or recognize features through the network (Ou et al., 2022). For instance, Neural network (Liu et al., 2020), Feature-based molecular networking (Nothias et al., 2020), Continuous-Time Network (Zhao and Park, 2015), Bayesian network (Albraikan et al., 2022), Complex network (Shi et al., 2022; Sun et al., 2022). Among them, indicators such as assortativity, centralization, density and transitivity in the complex network model can well reflect the cooperative tendency, important countries, market liquidity and group of the trade network. It has been used by many scholars to study the commodity trade relation structure (Sun, 2022). Therefore, this paper selects the complex network model to extract the characteristics of the trade network so as to quantify the evolution of the trade relation structure in different periods.

Up to now, many scholars have studied the influence of some factors on commodity price fluctuation (Balcilar and Ozdemir, 2019; Wu and Xu, 2021). Zhu et al. (2020) studies the correlation between RMB internationalization and nonferrous metal prices using a model called DCC-GARCH. Chai et al. (2021) used path analysis to extract the core driving factors affecting coal prices. Shao and Zhang (2020) used VAR model to study the spillover effect of crude oil price on the price of clean energy metals. Su et al. (2020) applied the TVP-SV-VAR model to investigate the effects of partisan conflict, dollar index and U.S. oil production on oil prices. Based on the impulse response function of VAR model, the response changes of other variables in a long time after a specific variable is shocked can be analyzed. This makes it easier to discuss the lasting influence on commodity prices after a shock to a commodity's trade relation structure. Therefore, this paper chooses the impulse response function to study the influence of the commodity trade relation structure changes on commodity prices.

In addition, in the copper industrial chain, upstream, midstream and downstream commodities are linked through close production relation, and their price fluctuations are also correlated (Guo et al., 2019; Mutreja et al., 2014). This means that changes in trade relation structure of commodities in one layer may influence prices in others (see Supplementary Fig. B.1). Therefore, this paper selects one representative commodity at each layer of the copper industrial chain. Firstly, the influence of a single commodity's trade relation structure on its price is studied. Then considering the association of the industrial chain, the influence of a certain commodity's trade relation structure on representative commodity prices at all layers of the industrial chain is studied. This will provide traders with a new direction of reference for their decision-making.

2. Data and methods

2.1. Data

One representative commodity of each layer in the copper industrial chain is selected as samples (Table 1). Since import data are usually more reliable from the perspective of tariffs, this paper selects import data for research (Shi et al., 2021). Import data for 32 years from 1988 to 2020 were downloaded from the UN Comtrade database according to the HS code.

2.2. Global trade network model

The vertices in the network represent the trading countries of the commodity. The arrow of the edge points to the importing countries. The weight of an edge is the trade quantities between the two countries. The trade network of commodity p in year t composed of vertices and edges as follows:

$$G_t = (V_{p,t}, E_{p,t}) \quad (1)$$

where $V_{p,t} = (v_{p,1,t}, v_{p,2,t}, v_{p,3,t}, \dots, v_{p,n,t})$, and n is the number of nodes that represent the number of trading countries of commodity p ; $E_{p,t} = (e_{p,1,t}, e_{p,2,t}, e_{p,3,t}, \dots, e_{p,m,t})$, and m is the number of edges that represent the total number of trade relations between countries. For each commodity p , a trade network is built each year from 1988 to 2020. Supplementary Fig. C.1 shows the trade networks taking the upstream commodity copper ore as an example.

2.3. The characteristics of commodity trade relation structure

In this paper, the following network topology and structural features are extracted to reflect the characteristics of the trade relation structure of copper commodities. Detailed explanations of each indicator are given in Appendix D.

(1) Assortativity: The scale of chain-like trade relations

The network assortativity is obtained by the following formulas (Newman, 2002, 2003):

$$\text{Assortativity} = \frac{1}{\sigma_{in}\sigma_{out}} \sum_{jk} jk (e_{jk} - q_j^{in} q_k^{out}) \quad (2)$$

Table 1
Information of representative commodities.

Layer	HS	Description
Upstream	2603	Copper ores and concentrates
Midstream	7403	Refined copper and copper alloys, unwrought
Downstream	7408	Copper wire

$$q_k^{in} = \sum_j e_{jk} \quad (3)$$

$$q_j^{out} = \sum_k e_{jk} \quad (4)$$

where σ_{in} , σ_{out} is the standard deviation of the distribution q_j^{in} , q_k^{out} respectively; e_{jk} is the probability that a randomly chosen directed edge is from a vertex of in-degree j to a vertex of out-degree k . The increase in assortativity means that there may be more chain-like trade relations. That is, the commodities may be imported several times before they are finally used. There are two main reasons for this phenomenon. One is that the importing country cannot directly import commodities from the main export countries, and the other is that the cost will be reduced in this way. Obviously, these reasons may cause the average price of commodities to fluctuate.

(2) Centralization: The status of important trading countries

The algorithm of network centralization is as follows (Liu, 2009):

$$Centralization = \frac{\sum_{i=1}^n (C_{max} - C_i)}{\max[\sum_{i=1}^n (C_{max} - C_i)]} \quad (5)$$

where C_{max} is the value of the maximum centralization of the vertex in the network and C_i is the centralization of vertex i . This represents the control power of the core vertices in the network (Freeman, 1979). In international trade, this reflects whether there are countries that plays important roles in the commodity trade. If the trade status of countries is unequal, the trade decisions of major exporting or importing countries may cause the fluctuation of commodity prices.

(3) Density: Market liquidity

The density algorithm is as follows (Wasserman and Faust, 1994):

$$Density = \frac{e_G}{e_{max}} \quad (6)$$

where e_G is the number of edges in the network and e_{max} is the number of possible edges in the network. This reflects the density of edges in the network. In international trade, more edges mean the establishment of more trade relations. This reflects the liquidity of commodity markets. High commodity market liquidity represents intense trade competition between countries. Countries in competition need to adjust commodity prices so that they may be in an advantageous position in the competition and may obtain the greatest benefits. Thus, commodity prices may fluctuate with changes in the commodity market liquidity.

(4) Transitivity: Trade group

In a weighted graph, transitivity is usually calculated as follows (Barrat et al., 2004):

$$T_i = \frac{1}{s_i(k_i - 1)} \sum_{j,h} \frac{w_{ij} + w_{ih}}{2} a_{ij} a_{ih} a_{jh} \quad (7)$$

$$Transitivity = \frac{\sum_i T_i}{n} \quad (8)$$

where T_i represents the transitivity of vertex i in the weighted network, s_i represents the weighted degree of vertex i , k_i represents the degree of the vertex i , w_{ij} represents the weight of the edge between vertex i and vertex j and a_{ij} are elements of the adjacency matrix. It is sometimes called the clustering coefficient, which is the degree of clustering of vertices in the network (Wasserman and Faust, 1994). In international trade, this represents the situation of the trade group. Countries in the trade group may reach some agreement on the price of commodities. On

the one hand, it can keep intra-group trade relatively stable. On the other hand, countries within the group can unite to compete with countries outside the group. These could also have an influence on commodity prices.

2.4. Variable selection

In order to quantify the influence of trade relation structure on commodity prices, this paper constructs VAR models. The models take price, assortativity, centralization, density, and transitivity as endogenous variables, and the trade quantity of commodities as control variables. The prices of the commodity are obtained by follows:

$$pri_{i,o,t} = \frac{V_{i,o,t}}{S_{i,o,t}} \quad (9)$$

$$pri_{i,t} = \frac{\sum_o^h pri_{i,o,t}}{h} \quad (10)$$

where, $pri_{i,o,t}$, $V_{i,o,t}$, and $S_{i,o,t}$ are the price, trade value and trade quantity respectively for trade relation o in year t . h represents the quantity of trade relation o . $pri_{i,t}$ represents the average annual price of commodity i , which is the average of prices for all trade relations o during year t . The trade quantity of commodities in year i ($V_{i,t}$) is the total trade quantity of all trade relations in that year, as shown below:

$$V_{i,t} = \sum_o^h V_{i,o,t} \quad (11)$$

To stabilize the model and make it persuasive, this paper first normalizes the original data to the range of [1, 2], and then takes the log-return of the normalized data (Formula (12) and (13)) as the final variables used.

$$y_{n,t} = 1 + \frac{y_{n,t} - y_{n,min}}{y_{n,max} - y_{n,min}} \quad t = 1, 2, \dots, T \quad (12)$$

$$y'_{n,t+1} = \ln(y_{n,t+1}) - \ln(y_{n,t}) \quad t = 1, 2, \dots, T \quad (13)$$

where $y_{n,t}$ and $y_{n,t+1}$ are the t th and $t+1$ th variables of variable n , respectively. $y_{n,min}$ is the minimum value of variable y_n and $y_{n,max}$ is the maximum value of variable n . $y'_{n,t+1}$ is the value of the $t+1$ th variable n that is finally used in the model. Table 2 shows the mean, median, maximum, minimum and standard deviation of each variable.

2.5. VAR model and impulse response function

The time series of trade relation structure for each commodity is obtained by constructing the annual trade network of each commodity. Then the commodity price time series corresponding to the four characteristic time series of each commodity are obtained. Considering the industrial chain, this paper discusses the influence of the change of commodity trade relation structure on commodity prices from the perspective of single commodity and industrial chain. Compared with the single commodity perspective, the industrial chain perspective considers the influence of the commodity trade relation structure on other commodity prices and the interaction between different commodity prices. The VAR model from the perspective of a single commodity is as follows (the descriptions of each variable are shown in Table 3):

$$\begin{cases} P_{p,t} = a_{1,1} TRS_{p,t-1} + a_{1,2} P_{p,t-1} + a_{1,3} TQ_{p,t-1} + \varepsilon_{P,p,t} \\ TRS_{p,t} = a_{2,1} TRS_{p,t-1} + a_{2,2} P_{p,t-1} + a_{2,3} TQ_{p,t-1} + \varepsilon_{TRS,p,t} \\ TQ_{p,t} = a_{3,1} TRS_{p,t-1} + a_{3,2} P_{p,t-1} + a_{3,3} TQ_{p,t-1} + \varepsilon_{TQ,p,t} \\ TRS \in \{Assortativity, Centralization, Density, Transitivity\} \\ p \in \{Upstream, Midstream, Downstream\} \end{cases} \quad (14)$$

Table 2
Descriptive statistics of the variables.

Variables	Mean	Median	Maximum	Minimum	Std. Dev.
Upstream Assortativity	-0.013	-0.005	0.279	-0.427	0.159
Midstream Assortativity	-0.011	-0.055	0.503	-0.345	0.179
Downstream Assortativity	0.012	0.004	0.257	-0.187	0.114
Upstream Centralization	0.009	0.005	0.155	-0.166	0.088
Midstream Centralization	0.019	0.008	0.284	-0.235	0.116
Downstream Centralization	0.009	0.001	0.362	-0.273	0.120
Upstream Density	0.021	0.017	0.474	-0.162	0.105
Midstream Density	0.022	0.010	0.390	-0.066	0.085
Downstream Density	0.018	0.023	0.550	-0.230	0.137
Upstream Transitivity	0.001	0.026	0.381	-0.414	0.149
Midstream Transitivity	0.012	0.000	0.455	-0.368	0.154
Downstream Transitivity	-0.001	0.024	0.248	-0.427	0.160
Upstream Price	0.007	0.014	0.240	-0.236	0.115
Midstream Price	0.011	-0.002	0.457	-0.337	0.137
Downstream Price	0.012	-0.005	0.350	-0.229	0.113
Upstream Trade Quantity	0.022	0.020	0.096	-0.017	0.021
Midstream Trade Quantity	0.022	0.029	0.091	-0.066	0.038
Downstream Trade Quantity	0.011	0.011	0.291	-0.304	0.085

Each trade relation structure feature of the commodity at each layer corresponds to one VAR model. Therefore, from the perspective of a single commodity, there are 12 models in total (Model1-12 in TableE.1). The VAR model from the perspective of industrial chain is as follows (the descriptions of each variable are shown in Table 3):

$$\begin{cases}
 UP_t = \beta_{1,1} TRS_{p,t-1} + \beta_{1,2} UP_{t-1} + \beta_{1,3} MP_{t-1} + \beta_{1,4} DP_{t-1} + \beta_{1,5} UTQ_{t-1} + \beta_{1,6} MTQ_{t-1} + \beta_{1,7} DTQ_{t-1} + \varepsilon_{UP,t} \\
 MP_t = \beta_{2,1} TRS_{p,t-1} + \beta_{2,2} UP_{t-1} + \beta_{2,3} MP_{t-1} + \beta_{2,4} DP_{t-1} + \beta_{2,5} UTQ_{t-1} + \beta_{2,6} MTQ_{t-1} + \beta_{2,7} DTQ_{t-1} + \varepsilon_{MP,t} \\
 DP_t = \beta_{3,1} TRS_{p,t-1} + \beta_{3,2} UP_{t-1} + \beta_{3,3} MP_{t-1} + \beta_{3,4} DP_{t-1} + \beta_{3,5} UTQ_{t-1} + \beta_{3,6} MTQ_{t-1} + \beta_{3,7} DTQ_{t-1} + \varepsilon_{DP,t} \\
 TRS_{p,t} = \beta_{4,1} TRS_{p,t-1} + \beta_{4,2} UP_{t-1} + \beta_{4,3} MP_{t-1} + \beta_{4,4} DP_{t-1} + \beta_{4,5} UTQ_{t-1} + \beta_{4,6} MTQ_{t-1} + \beta_{4,7} DTQ_{t-1} + \varepsilon_{DP,t} \\
 UTQ_t = \beta_{5,1} TRS_{p,t-1} + \beta_{5,2} UP_{t-1} + \beta_{5,3} MP_{t-1} + \beta_{5,4} DP_{t-1} + \beta_{5,5} UTQ_{t-1} + \beta_{5,6} MTQ_{t-1} + \beta_{5,7} DTQ_{t-1} + \varepsilon_{UTQ,t} \\
 MTQ_t = \beta_{6,1} TRS_{p,t-1} + \beta_{6,2} UP_{t-1} + \beta_{6,3} MP_{t-1} + \beta_{6,4} DP_{t-1} + \beta_{6,5} UTQ_{t-1} + \beta_{6,6} MTQ_{t-1} + \beta_{6,7} DTQ_{t-1} + \varepsilon_{MTQ,t} \\
 DTQ_t = \beta_{7,1} TRS_{p,t-1} + \beta_{7,2} UP_{t-1} + \beta_{7,3} MP_{t-1} + \beta_{7,4} DP_{t-1} + \beta_{7,5} UTQ_{t-1} + \beta_{7,6} MTQ_{t-1} + \beta_{7,7} DTQ_{t-1} + \varepsilon_{DTQ,t}
 \end{cases}
 \quad (15)$$

$TRS \in \{\text{Assortativity, Centralization, Density, Transitivity}\}$
 $p \in \{\text{Upstream, Midstream, Downstream}\}$

Similar to the perspective of a single commodity, each trade relation structure of the commodity at each layer corresponds to one VAR model. Therefore, there are 12 models in total from the perspective of the industrial chain (Model13–24 in Supplementary TableD.1). The difference is that the model from the industrial chain perspective takes into account the price and supply-demand of commodities at other layers. Since the trade relation structure of commodities changes frequently and the policy adjustments of countries are also very flexible, this article assumes that changes in commodity prices are only influenced by changes in the trade relation structure in the current year and the first year; that

Table 3
Nomenclature.

Variable	Description
$P_{p,t}$	The price of commodity p at period t
$TRS_{p,t}$	The trade relation structure of commodity p at period t
$TQ_{p,t}$	The trade quantity of commodity p at period t
ε	Residual
α	The coefficients of variables in a single commodity perspective model
UP_t	The price of upstream commodity at period t
MP_t	The price of midstream commodity at period t
DP_t	The price of downstream commodity at period t
UTQ_t	The trade quantity of upstream commodity at period t
MTQ_t	The trade quantity of midstream commodity at period t
DTQ_t	The trade quantity of downstream commodity at period t
β	The coefficients of variables in the industrial chain perspective model

is, the lag period of each variable is set to 1.

The impulse response function can analyze the dynamic response of other variables to the impulse of a specific variable. Then, the relations between variables are obtained from the perspective of the advance of the response period. This compensates for the VAR model. In this paper, the impulse response function based on VAR model is used to calculate the changes of commodity prices in 15 years after a trade relation structure changes. In the 0th period, a one-unit shock is given on the log-return of the corresponding trade relation structure variable. Then, influence of trade relation structure on commodity prices in the next 15 periods is obtained.

3. Results

3.1. Evolution of the commodities trade relation structure

This paper constructs trade networks for each commodity from 1988 to 2020. The vertices of the network are the importers and exporters of commodities. The edges of the networks represent trade relations. The arrows of the edge point from an exporting country to an importing country, and the weight of the edge is the trade quantity of the trade relations. Then the topological features which reflect the characteristics

of trade relation structure are extracted. Fig. 2 shows the evolution of commodities trade relation structure.

Except for the positive assortativity of the upstream trade network in 1988, the assortativity of all commodities in other years is negative. This shows that the copper commodity trade network is disassortative most of the time. That is, countries with similar external trade relations do not tend to establish connections. This means that in most cases, commodities are directly used after being imported. According to the law of comparative advantage, a country's comparative advantage in a certain industry is the root cause of establishing trading connections (Ricardo, 1821). When some countries have similar external trade relations with a

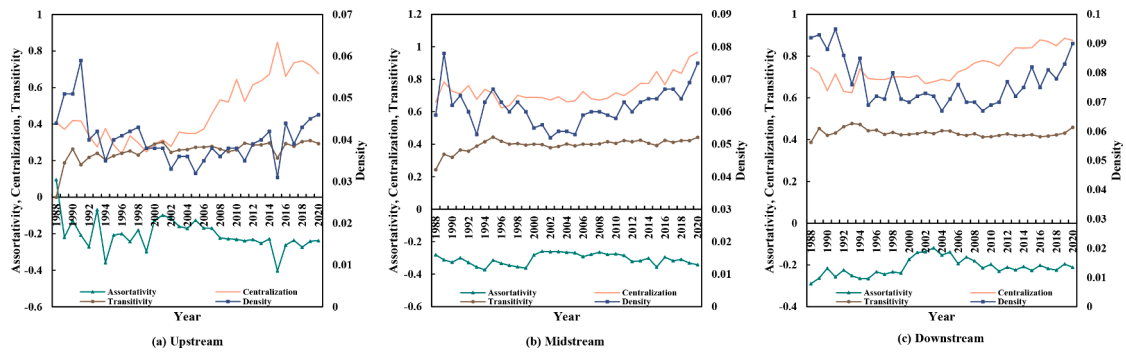


Fig. 2. Evolution of the commodities trade relation structure.

commodity, it means that they have a high probability of having a similar competitive advantage. Therefore, there is little tendency to establish trade relations between these countries. In addition, the volatility of the assortativity of the upstream trade network is greater than that of other layer. This may due to more countries being involved in the midstream and downstream commodities' trade, and the change in the trade relations among a few countries may have no obvious influence on the overall commodity trade relation structure.

Before 1996, the transitivity of networks slowly increased, and the density of networks slowly decreased. This shows the process of gradual improvement of the copper commodities trade relation structure. During this period, with the development of the economy, the number of trading countries gradually increased, and communities were gradually formed. The establishment of trade relations between countries has been relatively slow, so the density has slightly decreased.

The centralization of the midstream and downstream commodity networks is relatively stable, mostly between 0.6 and 0.9. This shows that the trade relation structure of midstream and downstream commodities has always been similar to a star shape. There are some

important core countries, that may have an important influence on the trade relation structure of commodities. The centralization of the upstream commodity network gradually increased after 2005. This means that after 2005, the status of important trading countries in the world's upstream copper trade is constantly improving.

It is worth noting that the upstream commodity trade relation structure, underwent major changes in approximately 2015. At the end of 2008, when the Federal Reserve embarked on quantitative easing, copper prices, led by futures, surged. The capital expenditure of international copper enterprises began to decrease in 2008, and reached a peak in 2013. Copper mines have a long lead time, taking 4–5 years from capex to capacity. This makes the expansion of copper capacity evident in the 2015 upstream trading network. In the 2015 network, 30 trading countries were not present in either 2014 or 2016. Only three of these countries are importing country, all others are exporting country, and most of the exporting countries export copper in small quantities.

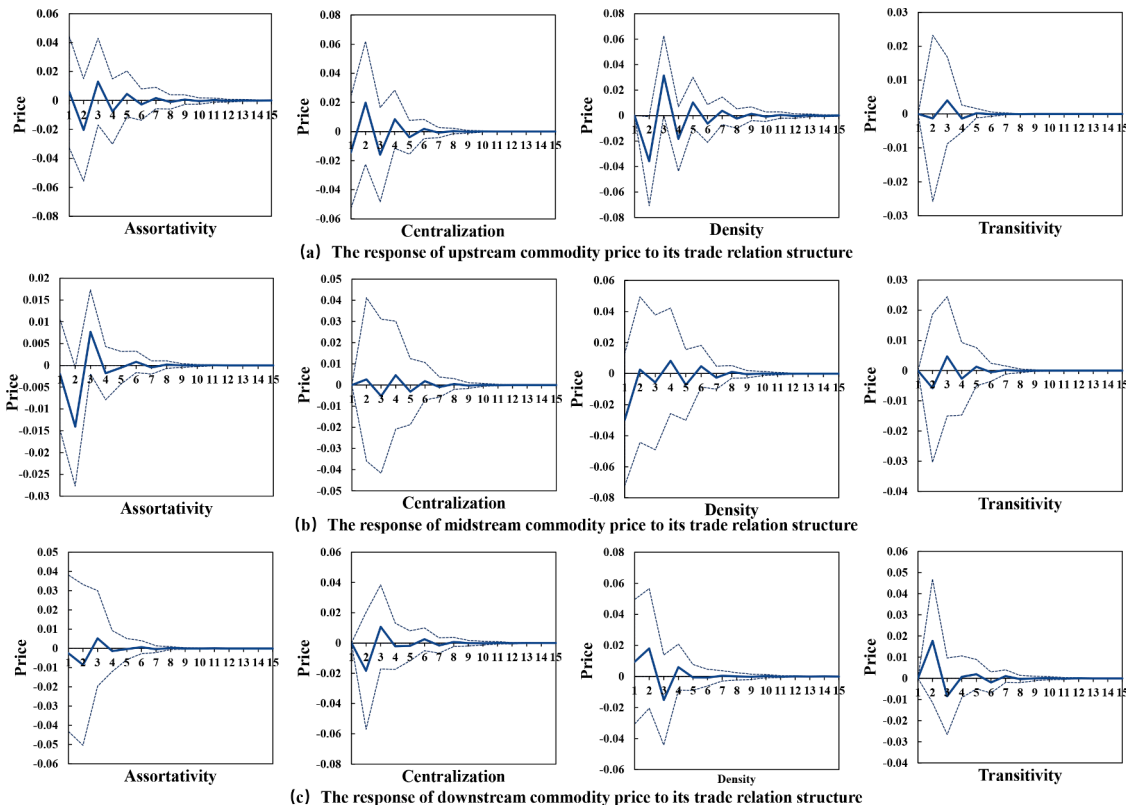


Fig. 3. The influence of trade relation structure on price from single commodity perspective.

3.2. The influence of trade relation structure on price: single commodity perspective

This paper first explores the influence of the copper trade relation structure on prices from the perspective of a single commodity. All the models obtained pass the AR root test; that is, all models are stable (Appendix F).

(1) Upstream commodity

The response of upstream commodity price log-return to its trade relation structure log-return is shown in Fig. 3(a). Among them, centralization reduces price log-return in the first year. The influence of other trade relation structure characteristics on price log-return mainly begins to manifest in the second year. Transitivity has little influence on price log-return. The influence of assortativity, centralization and density on upstream commodity price log-return decreases over time and basically disappears in the seventh year. Overall, trade relation structure only fluctuates the upstream commodity price log-return, but ultimately has little influence on its rise or fall.

Therefore, changes in the status of important countries should be reflected in a timely manner. Changes in the status of trading countries and the scale of chain-like trade relations will make prices fluctuate greatly. The influence of chain-like trade relations, important country status and market liquidity on price mainly concentrated in the first seven years after they change.

(2) Midstream commodity

The response of midstream commodity price log-return to its trade relation structure log-return is shown in Fig. 3(b). Among them, the price log-return of midstream commodities falls sharply in the first year after the density is shocked. Shocks to log-return of centralization and transitivity have little influence on prices. In the second to fourth years after the assortativity log-return shock, midstream commodity price log-

return will fluctuate greatly. On the whole, after the impulse on the trade relation structure of midstream commodities, log-return of assortativity and density will reduce the price log-return of midstream commodities, while other trade relation structure characteristics mainly cause price log-return fluctuations.

Therefore, the market liquidity of midstream commodities needs to be focused on and timely responded to. The influence of chain-like trade relations' scale on price is mainly concentrated in the second to fourth years after its change. Changes in the status of important countries and trade groups have little influence on midstream commodity price, which can be appropriately paid less attention to.

(3) Downstream commodity

The response of downstream commodity price to its trade relation structure is shown in Fig. 3(c). In the first year, density will make the price of downstream commodities rise and then produce fluctuations. The influence continued into the fifth year. The influence of other characteristics of trade relation structure on downstream commodity price begins in the second year. And the influence of these trade relation structure almost disappeared in the fourth year. This has a shorter duration than the influence on prices of trade relation structure in the upstream and midstream.

Thus, every change in the characteristics of the downstream commodity's trade relation structure of needs to be paid attention to by traders. Market liquidity has the biggest influence on its price in the first year, and the rest in the second to fourth years. The scale of chain-like trade relations causes the biggest fluctuation in midstream commodity price.

3.3. The influence of trade relation structure on price: industrial chain perspective

In the industrial chain, commodities at all layers are connected to each other through production relations, and the relations between

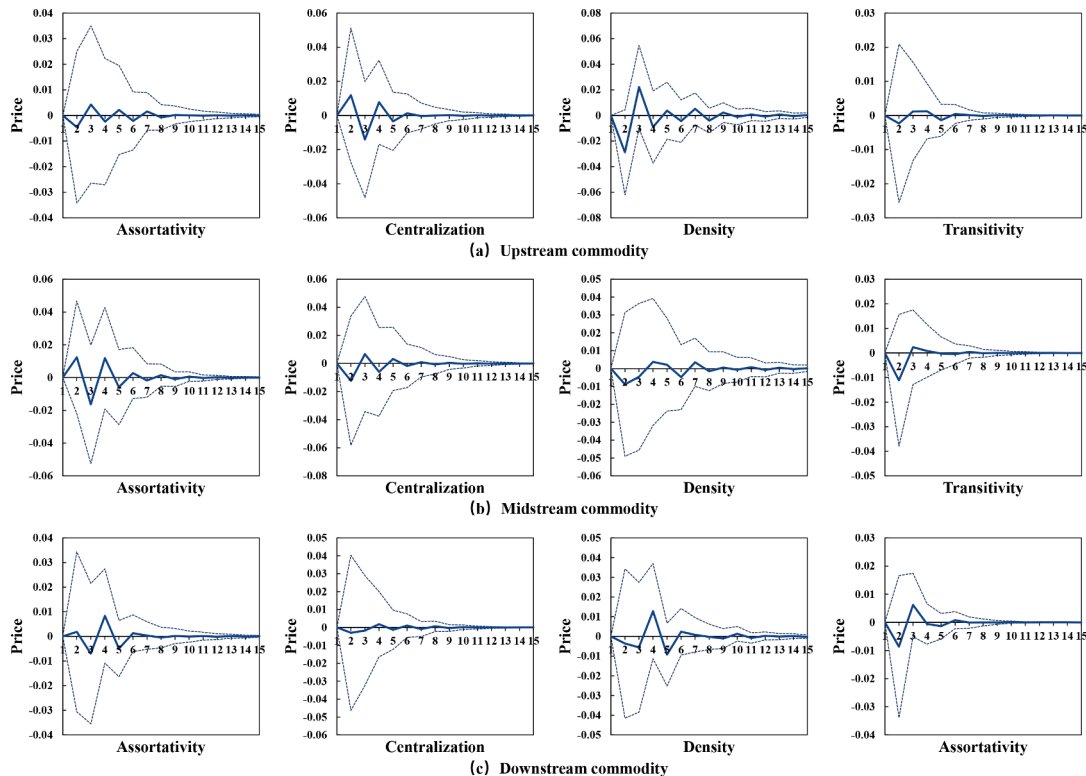


Fig. 4. The response of prices to the trade relation structure of upstream commodity.

supply and demand are close. For this reason, on the one hand, exporters and importers may make adjustments in advance due to changes in the trade relation structure of commodities at other layers. On the other hand, the prices of commodities are influenced not only by their own factors, but also by the factors of other commodities in the industrial chain. Therefore, compared with the influence of trade relation structure on its prices from a single commodity perspective, the following results consider the following two aspects: First, the trade relation structure of a commodity may not only influence its price, but also may influence the other commodity prices in the industrial chain. Second, the prices of commodities in the industrial chain can influence each other. All the models obtained pass the AR root test; that is, all models are stable (Appendix F).

(1) Upstream commodity

The response of commodity price to the trade relation structure of upstream commodity is shown in Fig. 4. After taking into account the industrial chain, the trade relation structure of upstream commodity has less influence on its price. At the same time, the change of upstream commodity trade relation structure will also cause the fluctuation of midstream commodity price. Among them, Transitivity will decrease midstream prices, with the influence concentrated in the first two years. And the influence of other trade relation structure characteristics will last until about the sixth year. Changes in upstream commodity centralization have little influence on downstream commodity price, while other trade relation structure characteristics have smaller influence on downstream commodity price.

This shows that the consideration of industrial chain moderates the influence of upstream commodity trade relation structure on its price. The scale of chain-like trade relations of upstream commodity has the greatest influence on midstream commodity price. Downstream commodity traders need not pay too much attention to the change of upstream commodity trade relation structure. Changes in the status of upstream important country have little influence on the downstream

commodity price.

(2) Midstream commodity

The response of commodity price to the trade relation structure of midstream commodity is shown in Fig. 5. After taking into account the industrial chain, the influence of midstream commodity's assortativity and centralization on its price decreases. Density does not influence its prices from the first year as it does from a single commodity perspective, but from the second year. On the whole, centralization has little influence on midstream commodity price, while other trade relation structure characteristics will make midstream commodity price fall after fluctuations. Midstream trade relation structure has smaller influence on upstream commodity price. Density has a greater influence on downstream commodity price.

This means that the status of important countries has little influence on midstream commodity prices. While the influence of other trade relation structure characteristics on midstream commodity prices is mainly concentrated in the first four years after their change. The market liquidity of midstream commodities has the greatest influence on the price of downstream commodities.

(3) Downstream commodity

The response of commodity price to the trade relation structure of downstream commodity is shown in Fig. 6. After taking into account the industrial chain, the influence of downstream commodity trade relation structure on its price changes less. The density of downstream commodities has larger influence on the upstream commodity price. The centralization of downstream commodity has a larger influence on the midstream commodity price.

Above all, the trade relation structure of downstream commodities has a relatively stable influence on its price, that is, it is less influenced by commodities in other layers of the industrial chain. Downstream commodity market liquidity has the greatest influence on upstream

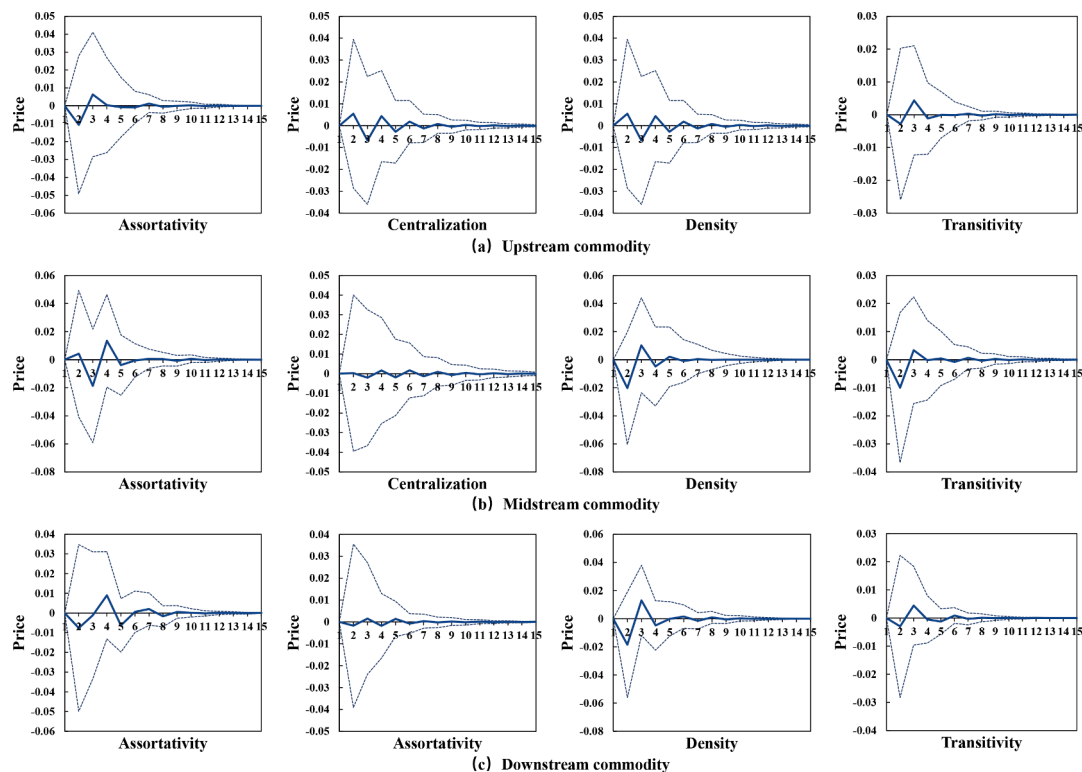


Fig. 5. The response of prices to the trade relation structure of midstream commodity.

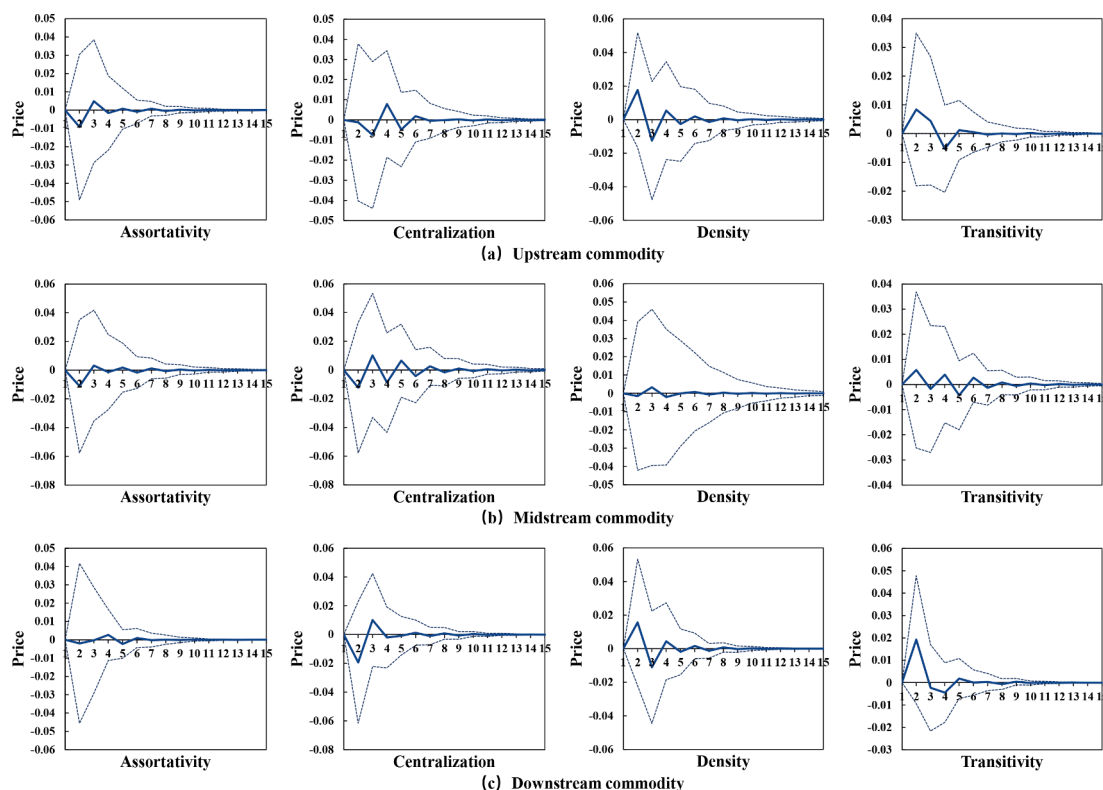


Fig. 6. The response of prices to the trade relation structure of downstream commodity.

commodity prices. The scale of the downstream commodity chain-like trade relations and the status of the important countries have greater influence on the price of midstream commodities. The influence of the scale of chain-like trade relations on midstream commodity price is mainly in the second year after its change.

4. Conclusions

Some evidence shows that the trade relation structure may have an influence on the copper price. Previous studies mainly considered the influence of agent supply-demand changes on commodity prices, while the research perspective of this paper is the influence of the supply-demand relation between the main agent changes on commodity prices, that is, the influence of the change of trade relation structure on the price. To confirm this idea, the representative commodities of each layer in the copper industrial chain are selected. Then this paper systematically combines the complex network theory and econometric methods and studies the influence of copper trade relation structure on price. The main conclusions are as follows.

The analysis of the evolution of commodity trade relation structure shows that the status of important countries in copper commodity trade is increasingly strengthened. There are some more stable groups among trading countries. Trading countries tend to trade directly, and countries with similar trade status are less likely to establish trade relations. The market liquidity of commodities fluctuates from year to year.

From the perspective of a single commodity, upstream commodity traders should pay more attention to changes in the status of important trading countries and in the scale of chain-like trade relations, because prices will fluctuate greatly after such changes. Midstream traders need to focus on and respond to liquidity issues in commodity markets in a timely manner. In addition, attention should also be paid to the scale of the chain-like trade relations of midstream commodities in the first 2–4 years. Downstream commodity traders need to pay attention to changes in the all characteristics of trade relation structure, but they only need to pay attention to the changes of the past four years.

However, due to the relatively complete copper industrial chain, it is necessary to explore the influence of commodity trade relation structure on commodity prices from the perspective of the industrial chain. From the perspective of industrial chain, upstream traders should pay more attention to the market liquidity of upstream and midstream commodities. Midstream traders should pay more attention to its market liquidity and the scale of chain-like trade relations. Downstream traders should pay more attention to the influence of the characteristics of the downstream commodity trade relation structure except the scale of chain-like trade relations and the market liquidity of midstream commodity.

This paper proposes a new perspective to discuss the influencing factors of copper price changes. This framework is applicable to any resource commodity as heavily dependent on trade as copper. Resource commodity traders can choose the appropriate decision-making perspective by comparing the difference of the influence of the trade relation structure on the price from the two perspectives. Then, the influence of the commodities' trade relation structure on commodity prices can be identified through the method of this paper to provide a reference for formulating short-term or long-term strategies. However, due to inadequate data length, only representative commodities are selected for analysis. Later, the existing work will be improved by seeking longer data or better models that can accommodate more short samples to include all commodities in the industrial chain.

CRedit authorship contribution statement

Yu Li: Methodology, Software, Writing – original draft. **Haizhong An:** Conceptualization, Funding acquisition. **Xiangyun Gao:** Conceptualization, Methodology. **Siyao Liu:** Writing – review & editing. **Qingru Sun:** Writing – review & editing. **Yiran Zhao:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.resconrec.2023.106933](https://doi.org/10.1016/j.resconrec.2023.106933).

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